Student Task Manager Application (Flutter with Hive)

Student Details

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Course: B.Tech CE
Subject: Flutter Development

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1. Introduction

The Student Task Manager Application is a Flutter mobile application developed to help students organize and manage their academic tasks efficiently. The application allows students to add tasks, track deadlines, and manage their study activities using a simple and user-friendly interface.

The application stores task information locally using the Hive database. This ensures that all tasks remain saved on the device even after the application is closed. The project demonstrates how Flutter applications can manage local data storage without relying on external servers or internet connectivity.

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2. Objectives

- To understand Flutter application development
- To implement local data storage using Hive database
- To demonstrate CRUD operations (Create, Read, Update, Delete)
- To develop a task management system for students

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3. Technologies Used

Flutter – Framework used to develop the mobile application

Dart – Programming language used in Flutter

Hive – Lightweight local database used for storing task data

GitHub – Used for version control and project hosting

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4. System Features

Add Task
Students can add a new task by entering task details such as title, subject, description, deadline, and priority.

View Tasks
All tasks are displayed in a list format so students can easily track their assignments and work.

Update Task
Students can edit and update task information whenever required.

Delete Task
Users can remove tasks that are no longer needed.

Task Status
Tasks can be marked as completed or pending.

Local Storage
All task data is stored locally using Hive database.

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5. System Workflow

Step 1
The user opens the Student Task Manager application.

Step 2
The user selects the option to add a new task.

Step 3
The user enters task details such as title, subject, deadline, and priority.

Step 4
The task data is stored locally in the Hive database.

Step 5
The user can view all saved tasks on the dashboard.

Step 6
The user can update or delete tasks when required.

Step 7
Tasks can be marked as completed once finished.

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6. Application Architecture

The application consists of the following components:

Main Application File (main.dart)
This file initializes the Hive database and starts the Flutter application.

Screens
Dashboard Screen – Displays task summary and task list
Add Task Screen – Allows users to add new tasks
Task Details Screen – Displays detailed information about a task
Completed Tasks Screen – Displays finished tasks

Database
Hive database is used to store and retrieve task records locally.

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7. Conclusion

The Student Task Manager application demonstrates how Flutter applications can store and manage task data locally using Hive database. The application allows students to organize their tasks, track deadlines, and manage their academic work efficiently. This project helps in understanding Flutter development, local database integration, and the use of GitHub for project version control.

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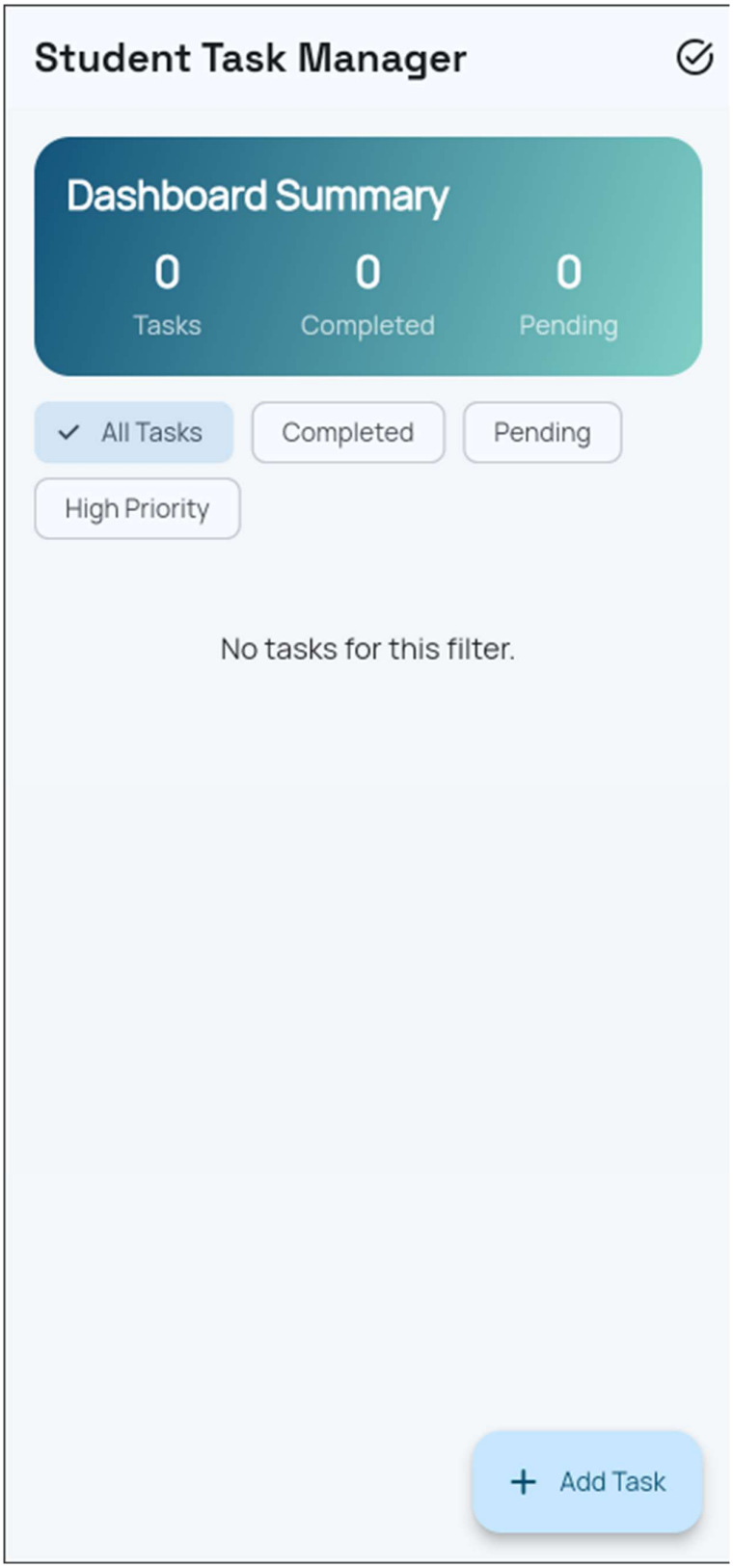
8. Screenshots


Student Task Manager
Dashboard Summary
0
Tasks
0
Completed
0
Pending
All Tasks
Completed
Pending
High Priority
No tasks for this filter.
Add Task

Add Task

Task Title

ala

Subject

mobile development

Description

api intigration

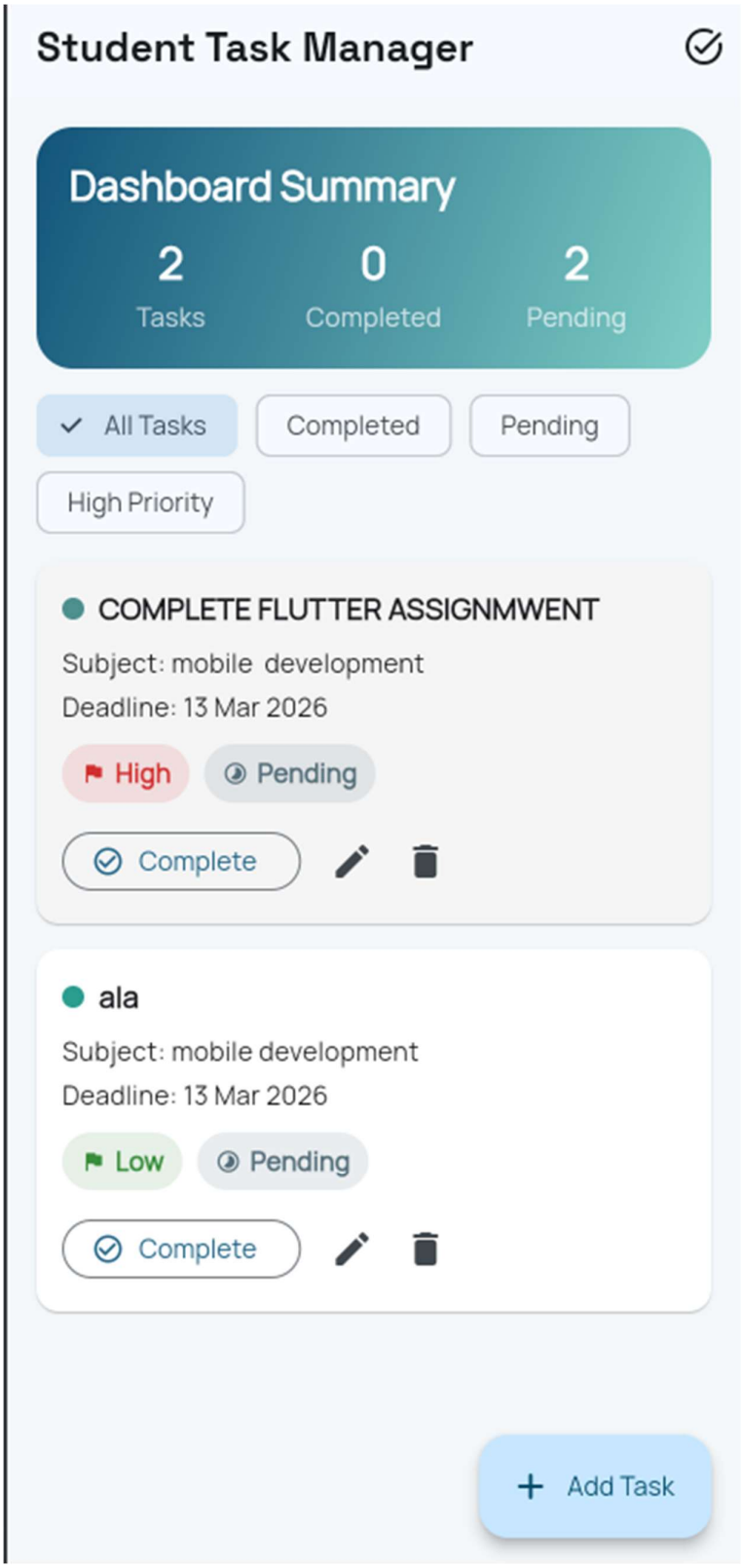
Deadline

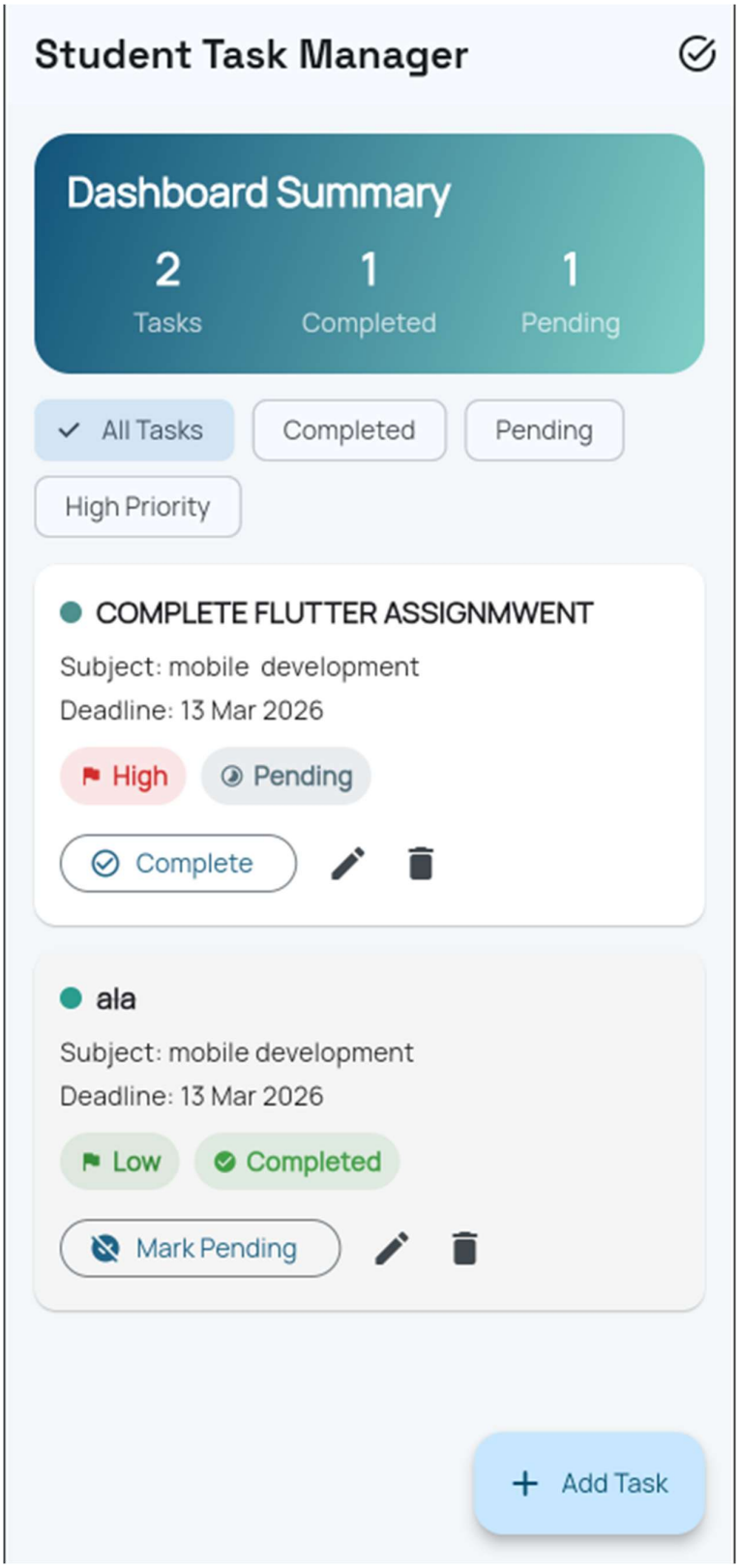
Fri, 13 Mar 2026

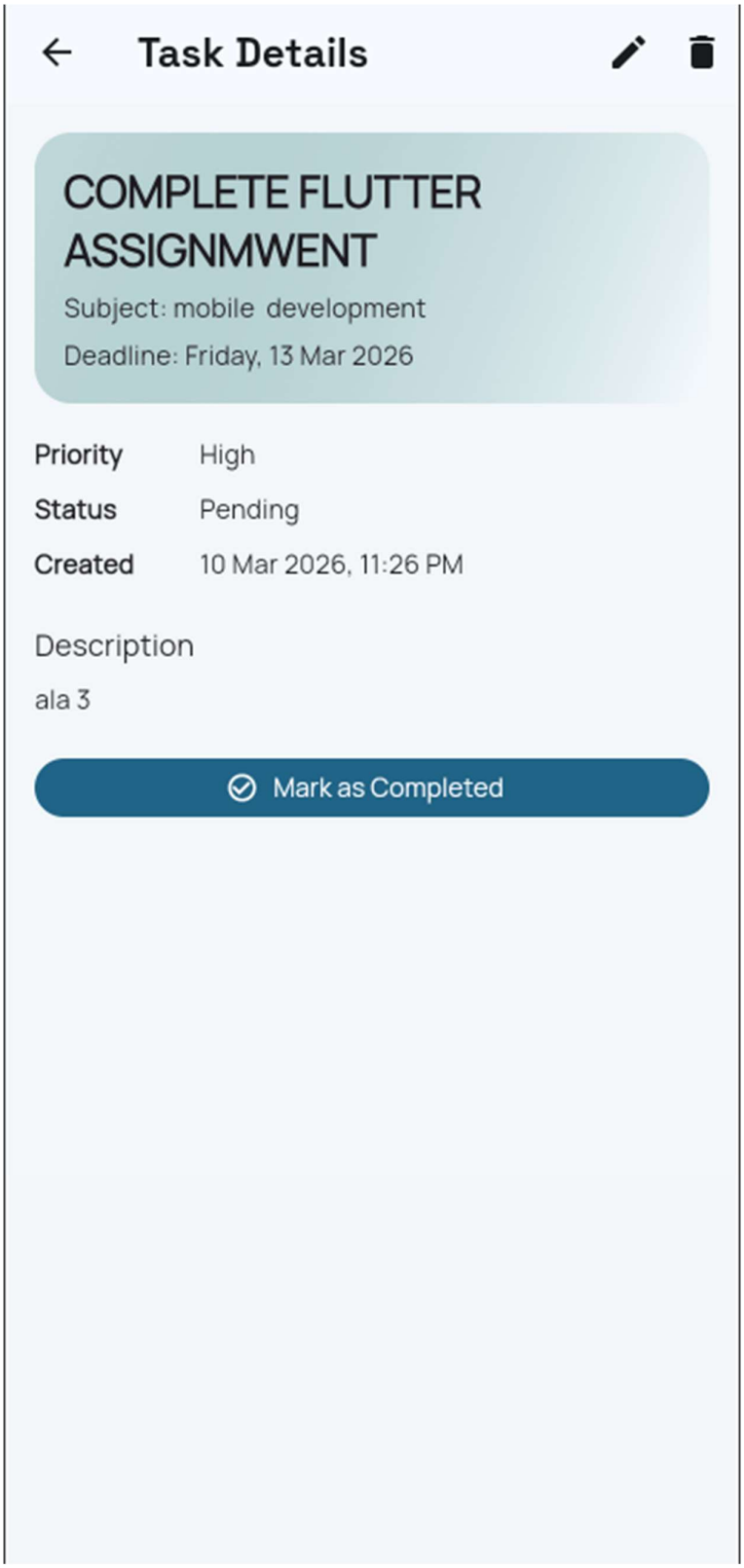
Priority

Low

+ Create Task


Student Task Manager
Dashboard Summary
2
Tasks
0
Completed
2
Pending
All Tasks
Completed
Pending
High Priority
COMPLETE FLUTTER ASSIGNMWENT
Subject: mobile development
Deadline: 13 Mar 2026
High
Pending
Complete
ala
Subject: mobile development
Deadline: 13 Mar 2026
Low
Pending
Complete
Add Task


Student Task Manager
Dashboard Summary
2
Tasks
1
Completed
1
Pending
All Tasks
Completed
Pending
High Priority
COMPLETE FLUTTER ASSIGNMWENT
Subject: mobile development
Deadline: 13 Mar 2026
High
Pending
Complete
ala
Subject: mobile development
Deadline: 13 Mar 2026
Low
Completed
Mark Pending
Add Task


Task Details
COMPLETE FLUTTER ASSIGNMWENT
Subject: mobile development
Deadline: Friday, 13 Mar 2026
Priority High
Status Pending
Created 10 Mar 2026, 11:26 PM
Description
ala 3
Mark as Completed